

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTETPE405E - PYTHON PROGRAMMING

Date: 11-07-2025

Time : 3 Hours]

[Total Marks : 60/33]

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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12

- ✓1 Python was developed as a successor to which language?
A) ABC language B) Perl C) Pascal D) Java
- ✓2 In Python, which of the following is used to execute code interactively?
A) Compiler B) IDE C) Terminal D) Python Shell (REPL)
- ✓3 What will be the output of print('Hello' + 'World')?
A) Hello World B) Hello+World C) HelloWorld D) Error
- ✓4 Which of the following is an immutable data type in Python?
A) List B) Dictionary C) Set D) Tuple
- ✓5 What will be the output of True and False?
A) True B) False C) 0 D) Error
- ✓6 Which operator is used to check membership in a collection?
A) == B) is C) in D) not
- ✓7 Which of the following denotes variable-length arguments in a function?
A) #args B) &args C) *args D) args[]
- ✓8 What is an anonymous function in Python called?
A) hidden function B) void function C) lambda function D) null function
- ✓9 Which method is called when an object is created?
A) start() B) new() C) create() D) init()
- ✓10 Which of the following is used to access class members from outside the class?
A) this B) self C) object D) def
- ✓11 Which module in Python provides functions to interact with the operating system?
A) sys B) os C) subprocess D) platform
- ✓12 Which module is used for mathematical operations like sin, cos, and log?
A) random B) math C) calc D) numpy

Q2. Solve the following.

- A) Discuss the major areas where Python is widely used. Why is it preferred over other languages in those areas? 6
- B) Write a short note on how Python scripts are executed. Include command-line and IDE-based execution methods. 6

Q3. Solve the following.

- A) Write a short note on arithmetic, relational, and logical operators in Python. 6
 C Include example expressions for each. 4
- B) Write a Python program to simulate a simple user-authentication system: 6
- Store usernames and passwords in a dictionary.
 - Ask the user to enter a username and password.
 - If matched, display a welcome message; else, allow 2 more attempts before exiting.

Q4. Solve Any Two of the following.

- A) Differentiate between import module, from module import, and import module as statements with examples. 6
- B) What are variable-length arguments in Python? Differentiate between *args and **kwargs with examples. 6
- C) Write a Python function calculate_total(price, quantity=1, discount=0) which: 6
- Returns the final price after applying the discount.
 - Uses default arguments for quantity and discount.
 - Accepts user input and prints the final bill.

Q5. Solve Any Two of the following.

- A) Write a short note on inheritance. 6
- B) Differentiate between an error and an exception in Python. Provide suitable examples of each. 6
- C) Write a Python program to demonstrate exception handling: 6
- Ask the user to enter two numbers.
 - Perform division and handle exceptions - ZeroDivisionError
 - Always display "Execution Completed" using finally.

Q6. Solve Any Two of the following.

- A) Why is testing important in programming? Explain unit testing in Python. 6
- B) Explain the use of the os module for interacting with the operating system. Mention any four commonly used functions. 6
- Write a Python program using the datetime module to: 6
- Ask the user to input a birthdate.
 - Calculate and display their current age in years, months, and days.

*** End ***